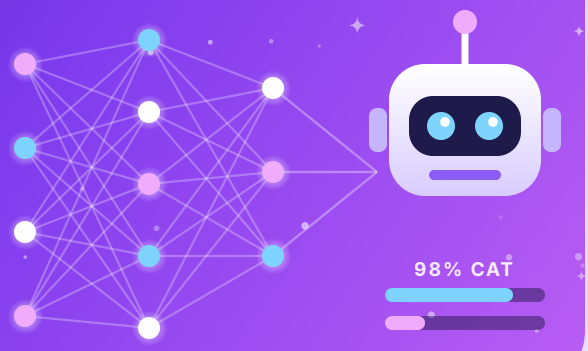
AGES 12–18 | MOST POPULAR | BASIC PYTHON NEEDED

CURRICULUM OVERVIEW

AI Builder Path

Build 3 Real AI Projects

A complete, project-driven journey from your first machine learning model to computer vision and large language model applications. Students don't just learn about AI — they write the code, train the models, and ship three working portfolio projects.



3 REAL AI PROJECTS
Models, vision apps and chatbots

4 TESTS ALONG THE WAY
3 course tests + 1 final test

3	24	28	12–18
Courses	Lessons	Sessions in total	Age range

WHO IT IS FOR
Our most popular program — for learners who already know basic Python

BEFORE YOU START
Basic Python knowledge needed

LEARNING PROGRESSION

01 Python and AI Foundations
Python skills, then a first machine learning model.

02 Computer Vision Projects
Apps that see: faces, emotions and objects in real time.

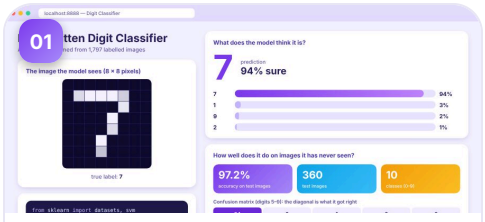
03 Language Models and Chatbots
Real LLM apps with memory, personality and a live interface.

Learn by building, not by watching
Every lesson pairs a real idea with a real, hands-on build. No shallow demos, no unfinished prototypes.

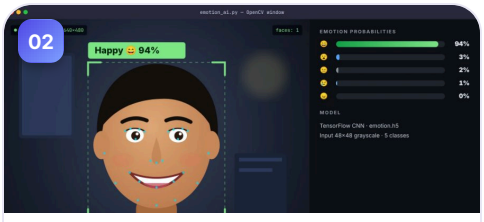
A portfolio, not just a certificate
Students finish with three working AI applications they can show, explain and keep building on.

Tested, so progress is real
Every course ends with a test, and a final test closes the program, so progress is proven, not assumed.

HANDS-ON PROJECTS STUDENTS BUILD



Handwritten Digit Classifier
Python · scikit-learn



Mood Detector AI
Python · TensorFlow · OpenCV



Recipe Chatbot
Python · OpenAI API · Streamlit

COURSE 01 · LESSONS 1–8

01 Python and AI Foundations

Write Python scripts and understand how machine learning really works

Python 3 | Jupyter Notebooks | scikit-learn | NumPy

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

The foundation of everything. With basic Python in hand, your child sharpens it for data work — variables, loops, functions and lists — then builds their first machine learning model. No fluff, just real code.

PORTFOLIO PROJECTS

- Number prediction model
- Simple image classifier
- Data pattern finder

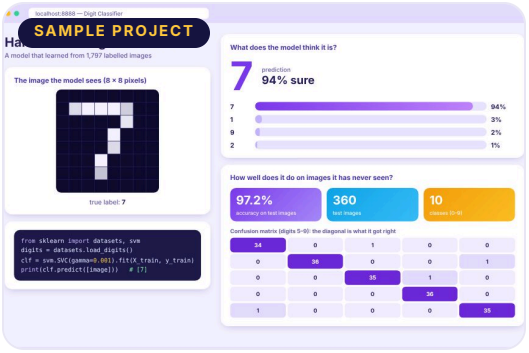
WHY IT MATTERS

Every machine learning system — from spam filters to recommendation engines — follows this same workflow: prepare data, train a model, test it honestly, and explain the result. This course is the foundation every later project builds on.

LESSON BREAKDOWN

1. Python Setup and First Program
2. Variables, Data Types, and Logic
3. Loops, Functions, and Modules
4. Working with Data and Lists
5. Introduction to Machine Learning Concepts
6. Building a First Prediction Model
7. Training, Testing, and Evaluating Models
8. Presenting Results to Non-Technical Audiences

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Handwritten Digit Classifier

Python · scikit-learn

A first image classifier: it learns from labelled examples, makes a prediction and reports how sure it is.

COURSE TEST

Python and AI Foundations — Course Test

Checks what you built and learned in Python and AI Foundations. Taken after lesson 8, before the next course begins.

30 multiple-choice | 10 true / false | 40 marks

COURSE 02 · LESSONS 9–16

02 Computer Vision Projects

Build apps that see, detect, and recognize the real world

Python · OpenCV · TensorFlow / Keras · Webcam API

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child builds apps that use a camera to detect faces, recognize objects, and read emotions — the same technology used in self-driving cars and medical diagnostics.

PORTFOLIO PROJECTS

- Real-time emotion detector
- Object recognition app
- Motion-activated security cam

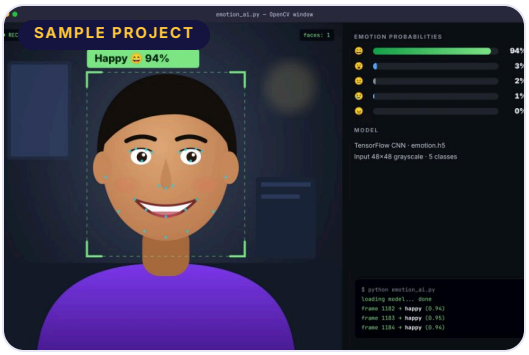
WHY IT MATTERS

Computer vision powers everyday technology students already use — unlocking a phone with a face, sorting photos, or a car spotting a pedestrian. Students learn how these systems actually work, including where they can fail.

LESSON BREAKDOWN

1. How Computers See — Pixels and Arrays
2. OpenCV Setup and First Image Processing
3. Face Detection with Pre-Trained Models
4. Emotion Recognition and Classification
5. Object Detection with YOLO
6. Real-Time Webcam Processing
7. Building a Full Computer Vision App
8. Deploying and Demoing Your Project

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Mood Detector AI

Python · TensorFlow · OpenCV

A webcam app that finds a face and guesses the emotion in real time, with the model's confidence on screen.

COURSE TEST

Computer Vision Projects — Course Test

Checks what you built and learned in Computer Vision Projects. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

COURSE 03 · LESSONS 17–24

03 Language Models and Chatbots

Build conversational AI apps using real LLM APIs

Python · OpenAI API · LangChain · Streamlit

3 REAL PROJECTS

1 COURSE TEST

OVERVIEW

Your child learns how ChatGPT actually works, then builds their own chatbot applications using the OpenAI API — complete with memory, personality, and real-world purpose.

PORTFOLIO PROJECTS

- Recipe chatbot with memory
- Study assistant bot
- Creative story generator

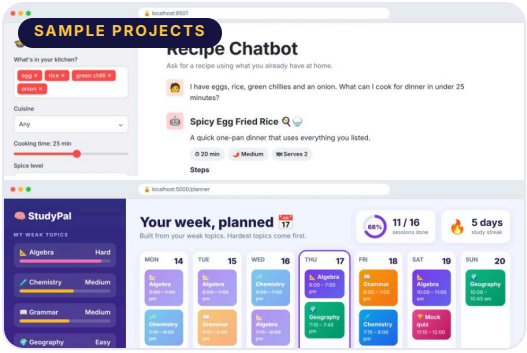
WHY IT MATTERS

Large language models now power search, customer support, and everyday writing tools. Students learn what's actually happening behind the chat window, then build and ship their own AI-powered application with a real interface.

LESSON BREAKDOWN

1. How Language Models Work (Transformers, Simplified)
2. OpenAI API Setup and Your First Call
3. Prompt Engineering Fundamentals
4. Building Conversational Memory
5. Giving Your Bot a Personality and Purpose
6. Connecting to External Data Sources
7. Building a UI with Streamlit
8. Deployment and Sharing

8 LESSONS — TEST UNLOCKS



HANDS-ON PROJECT

Recipe Chatbot · Smart Study Planner

Python · OpenAI API · Streamlit | Python · OpenAI API

Chatbots with memory and purpose: one suggests recipes from the ingredients at home, one plans a week of study.

COURSE TEST

Language Models and Chatbots — Course Test

Checks what you built and learned in Language Models and Chatbots. Taken after lesson 8, before the next course begins.

30 multiple-choice · 10 true / false · 40 marks

HOW STUDENTS ARE TESTED · 4 TESTS

Level-Up Checkpoints

In AI it is easy to run code you do not understand. The tests make sure students can explain how models are trained, evaluated and used — not just press run.



THE CHECKPOINT PATH — EVERY COURSE ENDS WITH A TEST, THEN ONE FINAL

Course 1
8 lessons
COURSE 1
Course 2
8 lessons
COURSE 2
Course 3
8 lessons
COURSE 3
FINAL

COURSE TEST · after every 8 lessons

Show what you built and learned

After every 8 lessons a course test checks the ideas and skills of that course, so each next course starts on solid ground.

- **30** multiple-choice questions — choose one of four answers
- **10** true / false questions
- **40** marks in total — 1 mark for every question

FINAL TEST · the finish line

One final test on the whole curriculum

One final test across machine learning, computer vision and language models: questions plus a small practical project.

- **25** multiple-choice questions
- **5** true / false questions
- **20** marks for one small practical project (about 15 minutes) that your teacher marks

A TASTE OF A TEST

SAMPLE · MULTIPLE-CHOICE

Why is some data kept aside as a test set?

- A To check the model on examples it has never seen ✓ CORRECT
- B To make training slower
- C To hide mistakes
- D To use less memory

SAMPLE · TRUE / FALSE

A model that is 99% accurate on its training data is always accurate on new data.

True | False ✓

Examples in the style of a real test — written for this document, not taken from one.

AFTER EVERY TEST

Saved as you go
Answers save automatically while a student works, so nothing is lost if the connection drops.

Checked by the teacher
The teacher reviews the report and shares the result with the student.

Fair for everyone
Questions and answer options appear in a different order for every student.

The Complete Journey

Across 3 courses, 24 lessons and 4 tests, students progress from their first machine learning model to building and deploying full AI applications — walking away with three real, working projects for their portfolio.

3 COURSES · 24 LESSONS

01 Python and AI Foundations
Number prediction model · Simple image classifier · Data pattern finder
8 lessons · + course test

02 Computer Vision Projects
Real-time emotion detector · Object recognition app · Motion-activated security cam
8 lessons · + course test

03 Language Models and Chatbots
Recipe chatbot with memory · Study assistant bot · Creative story generator
8 lessons · + course test

HOW STUDENTS ARE TESTED · 4 TESTS

COURSE TEST 1
Python and AI Foundations

COURSE TEST 2
Computer Vision Projects

COURSE TEST 3
Language Models and Chatbots

FINAL TEST
One final test on the whole curriculum
One final test across machine learning, computer vision and language models: questions plus a small practical project.

BY THE END, STUDENTS CAN

- Write real Python programs and explain how a machine learning model is trained, tested and evaluated.
- Build a computer vision application that detects faces and objects in real time from a webcam.
- Build and deploy a conversational AI application powered by a real large language model API.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.

Lumexa AI Builder · Ages 12–18